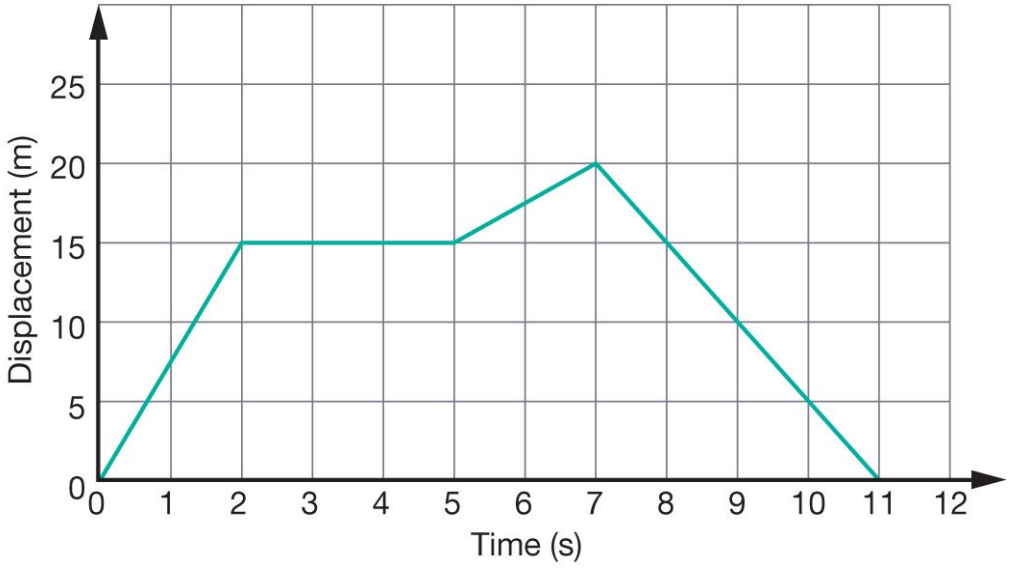
Chapter 8 Motion and energy

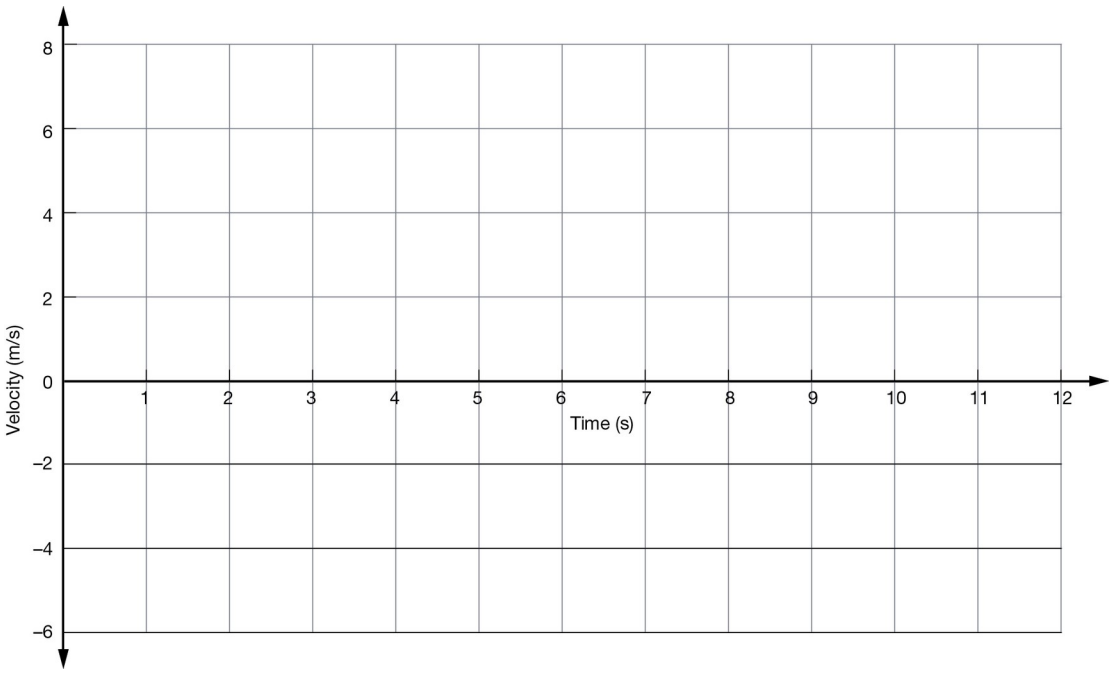
EXTRA WORKSHEET 2: Graphing motion

SCIENCE UNDERSTANDING **LOGICAL/MATHEMATICAL, VISUAL/SPATIAL**

1 The graph below shows the change in position of a stray cat as it runs down a straight laneway.



a **Construct** a velocity–time graph of the cat's motion on the axis below.



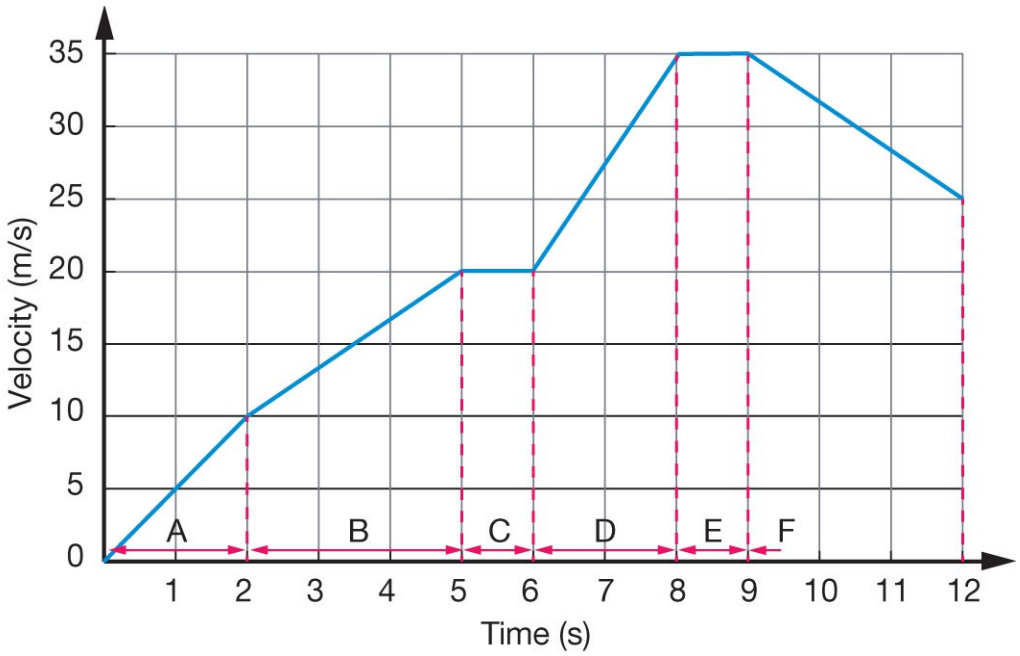
b **Describe** the motion of the cat as it runs along the laneway.

_____

_____

_____

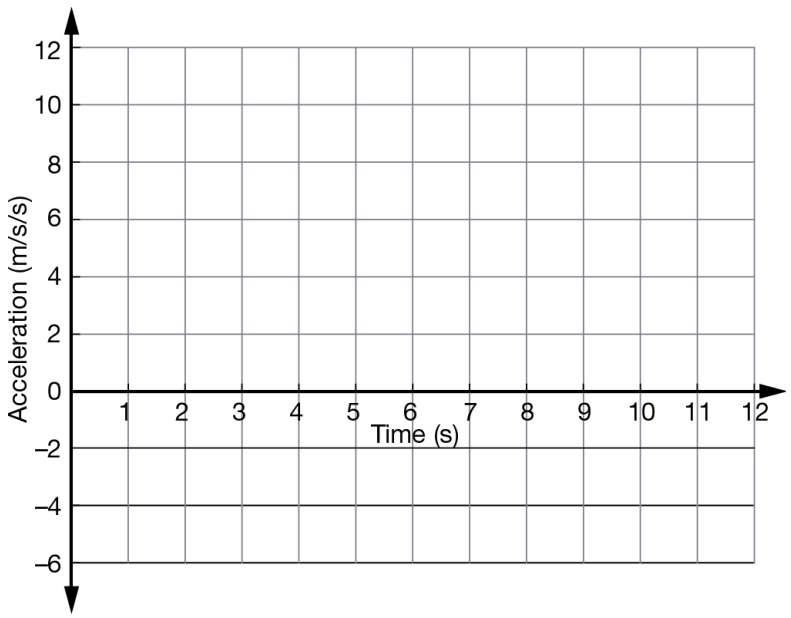
2 The graph below shows the motion of a cruise ship as it travels in a straight line route from the docks to its destination.



a **Calculate** the average acceleration of the ship in each of the sections marked A–F. Fill out your results in the table showing your working:

Section of motion	Average acceleration (m/s/s)
A	
B	
C	
D	
E	
F	

b **Use** your results from part **a** to **construct** an acceleration–time graph of the ship's motion on the axes below:



c **Identify** in which sections the ship is:

i speeding up: _____

ii slowing down: _____

iii travelling at a constant speed: _____

d **Calculate** the distance the ship sails in the first 5 seconds of its journey.